Basic Loop Questions

1. **Print numbers from 1 to 10 using a loop.**
2. **Print even numbers from 1 to 50 using a loop.**
3. **Print odd numbers from 1 to 50 using a loop.**
4. **Find the sum of the first `N` natural numbers using a loop.**
   (Input: `N`, Output: Sum of first `N` numbers)
5. **Find the factorial of a number using a loop.**
   (Input: `N`, Output: `N!` factorial)
6. **Print the multiplication table of a given number using a loop.**
   (Input: `N`, Output: Table of `N`)

Intermediate Loop Questions

7. **Reverse a number using a loop.**
   (Input: `12345`, Output: `54321`)
8. **Check if a number is a palindrome using a loop.**
   (Input: `121`, Output: `Palindrome`)
9. **Find the sum of digits of a number using a loop.**
   (Input: `123`, Output: `6`)
10. **Count the number of digits in a number using a loop.**
    (Input: `7865`, Output: `4` digits)
11. **Find the Greatest Common Divisor (GCD) of two numbers using a loop.**
12. **Print Fibonacci series up to `N` terms using a `for` loop.**
    (Input: `N=7`, Output: `0, 1, 1, 2, 3, 5, 8`)

Advanced Loop Questions

13. **Check if a number is prime using a loop.**
    (Input: `7`, Output: `Prime`)
14. **Print all prime numbers between 1 and 100 using a loop.**

15. **Check if a number is an Armstrong number using a `while` loop.**
    (Input: `153`, Output: `Armstrong Number`)
    ($153 = 1^3 + 5^3 + 3^3$)
17. **Find the sum of all prime numbers between 1 and 100 using a loop.**
18. **Convert a decimal number to binary using loops.**
    (Input: `10`, Output: `1010`)
19. **Write a program to reverse a given number.**

20) Write a program to find the largest digit in a given number.

Example:
Input: 341
Output: 4

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21) Write a program to find the smallest digit in a given number.

Example:
Input: 543
Output: 3

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22) Write a program to find the even digits in a given number.

Example:
Input: 34216
Output: 4, 2, 6

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23) Write a program to find the odd digits in a given number.

Example:
Input: 34216
Output: 3, 1

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24) Write a program to check if a number is a spy number or not.

Spy Number: A given number is a spy number if the summation and the product of its digits are equal.
Example:
Input: 123
Sum: 1 + 2 + 3 = 6
Product: 1 * 2 * 3 = 6
Output: Yes, 123 is a spy number.

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25) Write a program to check if the given number is a strong number or not.

Strong Number: A given number is a strong number if the sum of the factorials of its digits is equal to the number itself.
Example:
Input: 145
1! + 4! + 5! = 1 + 24 + 120 = 145
Output: Yes, 145 is a strong number.

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26) Write a program to check if the given number is a perfect number or not.

Perfect Number: A perfect number is a number whose sum of divisors (excluding the number itself) is equal to the number.
Example:
Input: 6
Factors: 1 + 2 + 3 = 6
Output: Yes, 6 is a perfect number.

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27) Write a program to check if the given number is a neon number or not.

Neon Number: A given number is a neon number if the sum of the digits of its square is equal to the number itself.
Example:
Input: 9
9 * 9 = 81
Sum of digits of 81 = 8 + 1 = 9
Output: Yes, 9 is a neon number.